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QatarEnergy

IS PROCESS HAZARD REVIEW (PHR) PROCEDURE

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1.0 PURPOSE

The purpose of this Document is to provide a systematic and consistent approach for management and operation of Process Hazard Reviews, PHRs, in accordance with the QatarEnergy/Idd El Shargi HSE Risk Management program and HSE Risk Management Standard.

2.0 SCOPE AND APPLICABILITY

This Procedure applies to all personnel within QatarEnergy/ Idd El Shargi who shall be involved in Process Hazard Review, and applies to Employees, contractors, and subcontractors, whether engaged directly by QatarEnergy or by others, who should perform work or services of any kind at any QatarEnergy/ Idd El Shargi facilities and/or locations. The Procedure shall be mandatory for HSE Process Hazard Review (PHR) activities carried out at QatarEnergy/ Idd El Shargi facilities.

3.0 ROLES AND RESPONSIBILITIES

TECHNICAL HSE ENGINEER

- Shall serves as change owner of the PHA-Pro8 software for PHR analysis.
- Shall Verify that the PHR process is being conducted in a satisfactory manner.
- Shall be responsible for control of : Reviews, Actions, Report outputs and Data Management.

HSE MANAGER

- Shall ensure that that all facilities within their sphere of responsibility have an effective written program for Process Hazard Review (PHR) activities and action tracking of the recommendations resulting from a PHR.

PHR FACILITATOR

- Shall be responsible for leading the PHR and preparing the documentation to support the results of the PHR.

PHR TEAM

- Shall be responsible for reviewing the hazards of the process and assessing the risk of identified hazards.

4.0 PROCEDURAL STEPS

The QatarEnergy/ Idd El Shargi Process Hazard Review (PHR) program shall address the following:

- Team Leadership.
- Team Composition.
- Facilities Technical Information.
- Scope Development.
- Documentation.
- Quality Control.
- Communication and Notification.
- Revalidation.

QatarEnergy/ Idd El Shargi shall conduct PHRs for new and existing Processes, to identify and understand the associated Hazards and HSE Risks.

For similar Processes with similar operating conditions, a single PHR should be representative of/ applicable to all. However, differences that exist in the operating conditions, such as the proximity to Receptors, shall be individually reviewed and assessed as part of the PHR.

PHRs shall evaluate credible and representative Event Scenarios.

It shall be accepted and understood that for Offshore operations and facilities, associated facility siting risks should have to be addressed on the basis that adequate administrative controls and similar mitigating factors shall be the only practicable method of Risk management.

For human factor considerations the PHR shall take into account access and egress, training content, safe work practices, control systems, human/machine interfaces, and human reliability.

Double/multiple jeopardy Event Scenarios are typically not considered credible or representative, but shall be evaluated if deemed credible by the PHR team.

4.1 CORE PHR PROCESSES

The table below illustrates the core PHRs utilized by QatarEnergy/ Idd El Shargi

ITEM	PROCESS	RECORD	OWNER	VALIDITY
1	Hazard Identification	HSE Risk Register	Technical HSE	Annual update
3	Hazard and Operability Study	Facilities HAZOP Files	Technical HSE, Process Engineering	5 years revalidation
4	Safety Integrity Level (SIL) reports	SIL Determination and Verification reports SIFs database	Technical HSE, Instrument Engineering	5 years revalidation
5	Quantitative Risk Assessment (QRA)	QRA report and Model	Technical HSE	5 years revalidation
6	Fire and Explosion Risk Assessment (FERA)	FERA report and Model	Technical HSE	5 years revalidation
7	Emergency Evacuation and Rescue Assessment (EERA)	EERA report	Technical HSE	5 years revalidation
8	Emergency System Survivability Analysis (ESSA)	ESSA report	Technical HSE	5 years revalidation

4.2 PHR SOFTWARE & ADMINISTRATION

QatarEnergy/Idd El Shargi employs PHA-Pro8 software, where relevant, for PHR analyses, provision of worksheet and review templates, report presentation and data management. QatarEnergy/Idd El Shargi - specific templates have been developed for a number of PHR processes, including, where relevant, the embedded QatarEnergy/ Idd El Shargi HSE Risk Matrix[1]. The change owner for this software in QatarEnergy/ Idd El Shargi shall be HSE Risk Engineering. Reviews, actions, report outputs and data management shall be controlled by the Technical HSE Engineer.

4.3 MOC-BASED PHR

4.3.1 MOC Level

The MOC committee determines the MOC Level in accordance with the criteria in MOC processes. This determination represents a form of initial hazard screening and, as such, a clear Statement of Requirement (SOR), should be presented to the MOC committee. Based on the MOC Level, the MOC committee then defines the PHR analyses to be conducted. Note that, irrespective of the MOC level, the MOC committee should mandate specific

[1] IS-HSE-PRC-301, Risk Evaluation and Response, Attachment 1.

analyses to suit the nature and complexity of the change – the intent is that the type and degree of PHR analyses should be selected based on the issues specific to that project/change[2]

4.3.2 Scope Change

Any reductions to the scope of the PHR program set by the MOC committee shall be justified and resubmitted to the committee for approval. Additional analyses, as required, should be actioned without MOC committee approval.

4.3.3 Core Processes

In addition to determining the required PHR analyses, the committee identifies any potential impact on 'core' processes.

4.4 REVIEW & ASSESSMENT

4.4.1 PHR Scope

PHRs shall be required for new and existing facilities under the QatarEnergy/ Idd El Shargi program. PHRs shall incorporate risk ranking of event scenarios and address the following elements:

- The hazards of the process.
- Conditioning factors for existing systems / equipment.
- Previous incidents which had a likely potential for catastrophic consequences.
- Engineering and administrative controls applicable to the hazards and their interrelationships such as appropriate application of detection methodologies to provide early warning of releases.
- Consequences of failure of engineering and administrative controls.
- Facility siting (of occupied buildings - permanent and portable).
- Human factors.
- A qualitative evaluation of a range of the possible safety and health effects of the failure of controls on facility employees.
- External events.

PHRs for these facilities shall also address the following items, as applicable:

- Safe separation distances (layout and spacing).
- Drainage and containment provisions related to flammable liquids.
- Toxic, fire and explosion hazards of the Process.
- Fire and gas detection.
- Fire protection.
- Accessibility to equipment and controls during normal and emergency operations.
- Electrical area classification.
- Waste collection/disposal.
- Potential offsite impacts.
- Process controls.

[2] QP-ISND-RM-HSE-304, Management of Change Procedure.

- Pressure relief Event Scenarios.
- Vent and blowdown discharge systems/locations.
- Utility systems.
- Storage systems and materials transfer systems.
- Corrosion & material Hazards, & any critical alloy circuits identified for testing.
- Maintenance requirements and equipment.
- Common Mode Failures, including failure of engineering & administrative controls.
- Knock-on Effects.
- Physical agents (noise, radiation, heat, stress, electrical shock hazards, etc.).
- Identification of previous incidents, including high potential near-misses, that presented a serious Hazard to multiple people, whether on or off site; and
- Modes of operation.

4.4.2 PHR Facilitators (Team Leaders)

Facilitator selection shall take into consideration the following skill sets:

- Effective Facilitation.
- Effective Communication.
- Organization.
- Attention to detail.
- Effective Listening.
- Conflict Resolution.
- Time Management.

The Facilitator shall ensure that the PHR is comprehensive and that the team's efforts shall be effectively directed towards the stated objective of identifying hazards and managing residual Risks and action items.

The Facilitator shall also provide an overview to the team members of the selected PHR methodology.

PHR Facilitators shall be formally trained in the methodology utilized (including the CME process[3]), and shall be qualified based on approved training, ongoing PHR leadership / participation and/or refresher training.

If the PHR Facilitator has not led a hazard review within the previous two years, then the leader shall undergo refresher training covering team leader requirements, PHR methodology and overview of the PHA-Pro software application.

4.4.3 Team Composition

In addition to the Facilitator, a PHR shall be performed by a team with expertise in the problem at hand, e.g. Engineering and Process operations. The team shall also include at least one employee who has experience and knowledge specific to the Process being evaluated.

[3] QP-ISND-RMG-HSE-301-50, CME Methodology Guideline.

Team members shall also be knowledgeable in the specific PHR methodology being employed.

Dependant on the complexity and scope of the process being evaluated, the following persons shall also be considered for team composition:

- PHR Scribe.
- Process Engineer.
- Operations Representative.
- Engineering Representative – Technical Authority or competent delegate.
- Maintenance Engineer and/or Technician.
- Computer Control / Automation Representative.
- HSE Representative.
- RE Team Member - required for major projects.
- AI / MI Representative.
- Others (as determined by the PHR Facilitator).

5.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Risk Evaluation & Response Procedure	IS-HSE-PRC-301	Internal	Upwards
Management of Change (MOC) Procedure	QP-ISND-RM-HSE-304	Internal	Sideways
Operations & Maintenance (O&MP) Procedure	QP-ISND-RM-HSE-306	Internal	Sideways
Pre-Startup Safety Reviews (PSSR) Procedure	IS-HSE-PRC-307	Internal	Sideways
Task Risk Assessment (TRA) Guideline	IS-HSE-GDL-315	Internal	Sideways
What If Analysis Guideline	IS-HSE-GDL-316	Internal	Sideways
Design Safety Analysis (DSA) Guideline	IS-HSE-GDL-317	Internal	Sideways

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Hazard & Operability (HAZOP) Review Guideline	IS-HSE-GDL-318	Internal	Sideways
Safety Integrity Level (SIL) Guideline	IS-HSE-GDL-319	Internal	Sideways
Evacuation, Escape & Rescue Analysis (EERA) Guideline	IS-HSE-GDL-320	Internal	Sideways

6.0 DEFINITIONS

TABLE 2: KEY TERMS DEFINED

Term	Definition
PROCESS HAZARD ANALYSIS	An organized, thorough and systematic approach to identify and evaluate hazards associated with chemical processes and operations to enable their control.
HAZOP	The application of a systematic critical examination, to the process and engineering intentions, of a facility to assess the hazard and operability potential of mal-operation or malfunction of individual items of equipment and the consequential effects on the facility as a whole.
What-If-Analysis	What–If Study is a structured brainstorming method of determining what can go wrong and judging the likelihood and consequences. The answers to these questions form the basis for making judgments regarding the acceptability of those risks and determining a recommended course of action for those risks judged to be unacceptable.
MANAGEMENT OF CHANGE (MOC)	MOC is a formal process that systematically manages permanent, temporary or emergency changes to the organizations, operations, process equipment, technologies, procedures or chemicals that could impact the original design, operations and maintenance criteria, and/or environmental performance.

APPENDIX

APPENDIX A – REVISION HISTORY LOG

TABLE 3: REVISION HISTORY LOG

Revision Number: 01 **Date: 06/12/2023**

Item Revised	Revision Description	Page No.
All	Document simplified to new format as part of QMR initiative.	All
Remarks:		

APPENDIX B - LIST OF ABBREVIATIONS

TABLE 4: LIST OF ABBREVIATIONS

Abbreviation	**Description**
DSA	Design Safety Analysis
EERA	Evacuation, Escape & Rescue Analysis
FERA	Fire and Explosion Risk Assessment
HAZOP	Hazard & Operability Review
MOC	Management of Change
PHA	Process Hazard Analysis
PHR	Process Hazard Review
QRA	Quantitative Risk Assessment
SIL	Safety Integrity Level